245122: volcaniclastic sandstone, Carbine mine

('Paradigm sandstone', Black Flag Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Kalgoorlie (SH 51-9), Davyhurst (3037)
MGA Zone 51, 301832E 6627510N

Sampled on 22 September 2020

This sample was collected from the 263.90 – 265.72 m depth interval of drillcore PDDD18003, drilled in 2018 by Northern Star Resources Limited at their Carbine prospect. The drillhole is located on Mount Burges Station, about 7.8 km east-southeast of Rowles Lagoon, 5.5 km southwest of Brown Lagoon, and 0.6 km north-northwest of the Carbine mine open pit.

Geological context

The unit sampled is a felsic volcaniclastic sandstone, known informally as the Paradigm sandstone, and assigned to the Black Flag Group (Morris, 1998; Hall et al., 2021). The 2692–2655 Ma Black Flag Group of the southwestern Eastern Goldfields Superterrane is an extensive northwest-trending unit, consisting of felsic metavolcaniclastic and metasedimentary rocks with subordinate felsic and mafic metavolcanic rocks (Squire et al., 2010). The Black Flag Group overlies the 2753–2680 Ma Kalgoorlie Group, is unconformably overlain by the 2664–2655 Ma Merougil Formation (Hall et al., 2021), and in part is intruded by the c. 2680 Ma Golden Mile Dolerite (Rasmussen et al., 2009). A porphyritic andesite of the Black Flag Group, collected about 36 km to the east-southeast, yielded an igneous crystallization age of 2690 ± 7 Ma (GSWA 211051, Lu et al., 2017).

Petrographic description

The sample is a foliated volcaniclastic sandstone (Fig. 1), consisting of about 60–70% microcrystalline material and altered plagioclase, 20–25% quartz, 7–8% biotite, 5–7% sulfide minerals, and minor chlorite, carbonate minerals and zircon. The texture of this rock has been substantially modified by deformation, recrystallization and phyllic alteration. The microcrystalline material displays few grain shapes or internal textures although rare shapes up to 2 mm long may be volcanic-derived clasts of felsic groundmass material. Plagioclase grain shapes are not well preserved, although a few equant to tabular grains are up to 1 mm in size. Plagioclase has been albitized and some grains are altered to sericite and carbonate. Detrital quartz grains constitute about 10% of the sample and a similar amount occurs in recrystallized lenses. Quartz grains are up to 2 mm in size and rare preserved embayments suggest volcanic provenance. Quartz grains are variably strained, and some are fractured or slightly recrystallized at their margins. About half the quartz is recrystallized and forms well-aligned lenses up to 7 mm long, composed of polygonal quartz with intergrown carbonate and sulfide minerals. Secondary biotite forms anhedral, irregular grains up to 1 mm in size, and crudely aligned subhedral laths up to 0.3 mm long. Some biotite is slightly altered to chlorite. Sulfide minerals include irregular, anhedral pyrrhotite up to 1 mm in size and anhedral arsenopyrite up to 0.1 mm across. The sandstone is interpreted as relatively poorly sorted and derived from a felsic volcanic source. Minor variation in grain size is interpreted as primary layering and is parallel to the foliation defined by quartz lenses and biotite. The sample has been partially recrystallized at the biotite zone of greenschist facies.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and mainly subhedral. The crystals are up to 350 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 June and 8 July 2021, using SHRIMP-B. Analyses 1.1 to 18.1 (spot numbers 1–18) were obtained during the first session, together with 10 analyses of the M257 standard, of which seven analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.29% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.53% (1σ). Analyses 19.1 to 7.2 (spot numbers 19–26) were obtained during the second session, together with three analyses of the M257 standard, of which two analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.27% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-six analyses were obtained from 22 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3, 4).

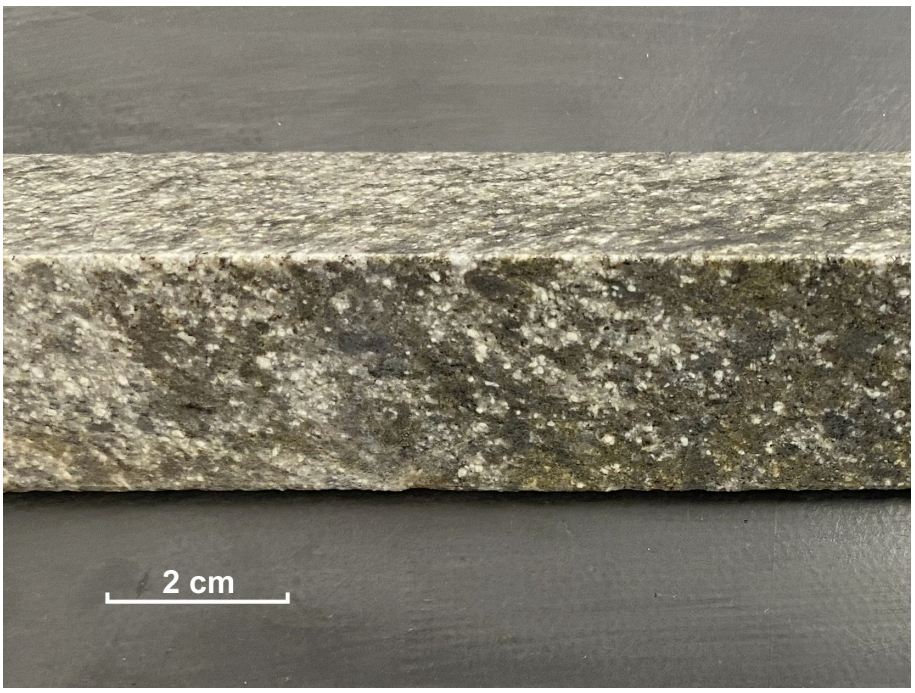

Figure 1. Drillcore image (wet surface) of sample 245122: volcaniclastic sandstone, Carbine mine

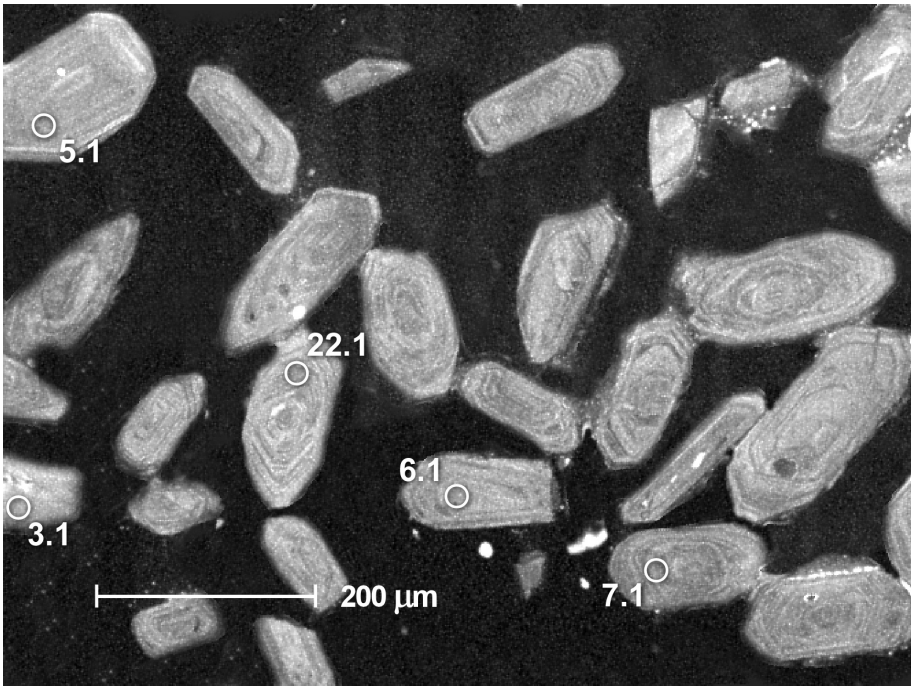

Figure 2. Cathodoluminescence image of representative zircons from sample 245122: volcaniclastic sandstone, Carbine mine. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 245122: volcaniclastic sandstone, Carbine mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	13	13.1	337	306	0.91	0.023	1.978	0.030	0.18426	0.00065	1.979	0.030	0.18405	0.00066	2637	34	2690	6	2.0
I	16	16.1	431	175	0.41	0.023	2.023	0.030	0.18473	0.00059	2.023	0.030	0.18452	0.00060	2589	33	2694	5	3.9
I	3	3.1	126	120	0.95	-0.016	2.011	0.034	0.18439	0.00107	2.011	0.034	0.18453	0.00108	2602	37	2694	10	3.4
I	8	8.1	337	147	0.43	0.019	2.057	0.031	0.18484	0.00068	2.057	0.031	0.18467	0.00069	2554	33	2695	6	5.3
I	26	7.2	652	298	0.46	0.023	1.975	0.014	0.18510	0.00042	1.975	0.014	0.18490	0.00043	2641	16	2697	4	2.1
I	15	15.1	271	147	0.54	0.057	2.016	0.031	0.18559	0.00078	2.017	0.031	0.18509	0.00080	2596	34	2699	7	3.8
I	25	4.2	402	352	0.88	0.000	2.100	0.016	0.18509	0.00053	2.100	0.016	0.18509	0.00053	2510	16	2699	5	7.0
I	19	19.1	383	75	0.20	-0.004	1.977	0.015	0.18556	0.00055	1.977	0.015	0.18559	0.00055	2639	17	2704	5	2.4
I	11	11.1	677	327	0.48	0.014	1.935	0.028	0.18622	0.00052	1.935	0.028	0.18609	0.00052	2686	33	2708	5	0.8
I	23	9.2	267	181	0.68	0.027	2.035	0.016	0.18637	0.00065	2.036	0.016	0.18612	0.00066	2576	18	2708	6	4.9
I	14	14.1	507	79	0.16	0.012	1.968	0.029	0.18629	0.00056	1.969	0.029	0.18618	0.00056	2648	33	2709	5	2.2
I	7	7.1	302	173	0.57	0.041	2.008	0.031	0.18734	0.00072	2.009	0.031	0.18698	0.00073	2604	34	2716	6	4.1
P	10	10.1	554	235	0.42	0.058	2.147	0.032	0.17825	0.00052	2.148	0.032	0.17773	0.00054	2464	31	2632	5	6.4
P	18	18.1	798	734	0.92	0.006	2.135	0.031	0.17793	0.00045	2.135	0.031	0.17788	0.00046	2476	31	2633	4	6.0
P	24	14.2	589	175	0.30	0.015	2.032	0.015	0.18008	0.00046	2.032	0.015	0.17995	0.00046	2579	16	2652	4	2.8
P	9	9.1	495	247	0.50	0.004	2.058	0.031	0.18220	0.00054	2.058	0.031	0.18216	0.00054	2553	32	2673	5	4.5
P	4	4.1	609	340	0.56	0.031	2.018	0.030	0.18311	0.00048	2.019	0.030	0.18283	0.00049	2594	32	2679	4	3.2
P	2	2.1	567	287	0.51	0.014	1.992	0.029	0.18301	0.00051	1.992	0.029	0.18289	0.00052	2623	33	2679	5	2.1
P	5	5.1	265	138	0.52	0.037	1.973	0.030	0.18404	0.00074	1.974	0.030	0.18371	0.00075	2642	34	2687	7	1.7
P	1	1.1	505	312	0.62	0.107	2.006	0.030	0.18467	0.00056	2.009	0.030	0.18371	0.00059	2605	33	2687	5	3.1
D	12	12.1	2002	230	0.11	0.014	3.128	0.045	0.12478	0.00050	3.129	0.045	0.12466	0.00050	1788	23	2024	7	11.7
D	6	6.1	2031	795	0.39	0.028	2.719	0.039	0.14092	0.00050	2.720	0.039	0.14066	0.00050	2018	25	2235	6	9.7
D	17	17.1	1676	511	0.31	0.141	2.898	0.043	0.14288	0.00033	2.902	0.043	0.14163	0.00036	1909	25	2247	4	15.1
D	20	20.1	927	392	0.42	0.042	2.592	0.022	0.15490	0.00126	2.593	0.022	0.15453	0.00126	2103	16	2397	14	12.3
D	21	21.1	868	505	0.58	0.015	2.336	0.016	0.16597	0.00035	2.336	0.016	0.16583	0.00036	2297	14	2516	4	8.7
D	22	22.1	706	480	0.68	0.005	2.437	0.017	0.16966	0.00039	2.437	0.017	0.16962	0.00039	2216	13	2554	4	13.2

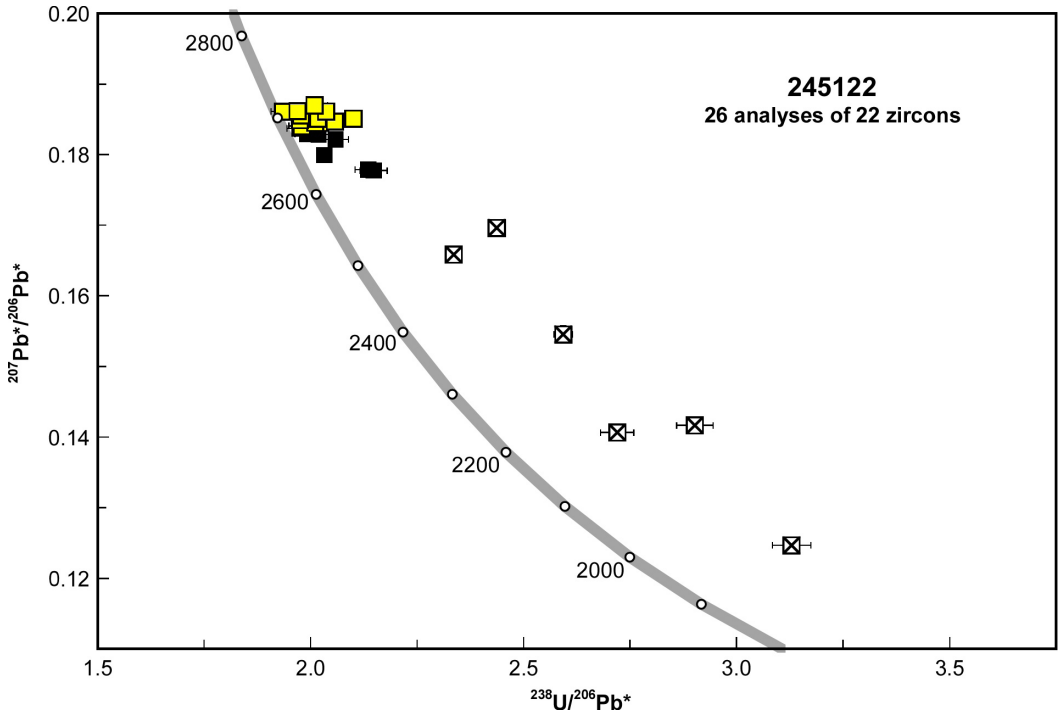

Figure 3. U–Pb analytical data for sample 245122: volcaniclastic sandstone, Carbine mine. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >7%)

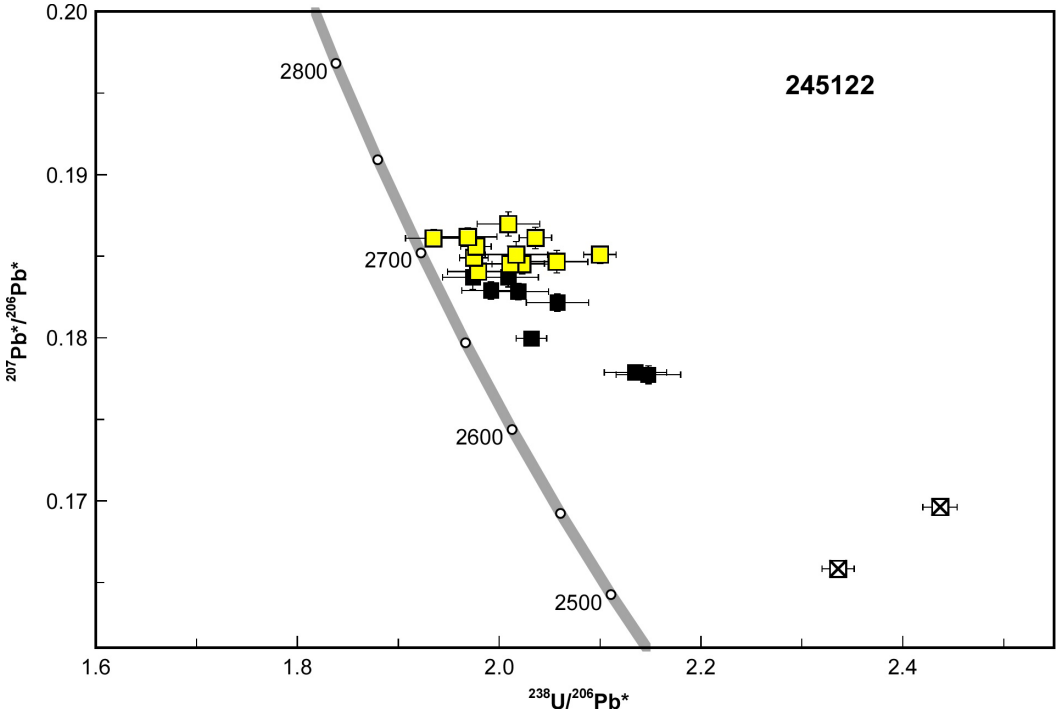

Figure 4. Expanded view of U–Pb analytical data for sample 245122: volcaniclastic sandstone, Carbine mine. Symbols as in Figure 3

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Six analyses are >7% discordant. The dates obtained from these six analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 20 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 12 analyses of 11 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2701 ± 5 Ma (MSWD = 1.8).

Group P comprises eight analyses of eight zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2687–2632 Ma.

The date of 2701 ± 5 Ma for the 12 analyses in Group I is interpreted as the igneous crystallization age of the volcanic rock. The dates of 2687–2632 Ma for the 10 analyses in Group P are interpreted to reflect loss of radiogenic Pb.

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Wingate, MTD, Fielding, IOH, Lu, Y and Smithies, RH 2021, 245122: volcaniclastic sandstone, Carbine mine; Geochronology Record 1837: Geological Survey of Western Australia, 5p.

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